

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – 2023

Course: B. Tech.

Branch: IT

Semester: V

Subject Code & Name: BTITC501 Database Management Systems

Max Marks: 60

Date:01-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Define i. Candidate key ii. Primary key iii. Superkey List the main functions of a database administrator.	Knowledge/ Comprehension	6
B) Discuss the main characteristics of the database approach and specify how it differs from the traditional file system.	Analysis	6
C) Explain different types of Attributes with examples.	Application	6
Q.2 Solve Any Two of the following.		12
A) Draw an E-R diagram of library management system considering issue and return, fine calculation facility, also show primary key, weak entity and strong entity.	Synthesis	6
B) Write SQL queries for following. Consider table stud (roll no, name, sub1, sub2, sub3) i. Display names of student who got minimum mark in sub1. ii. Display names of students who got above 40 marks in sub2. iii. Display count of Students failed in sub2. iv. Display average marks of sub1 of all students. v. Display names of students whose name start with 'A' by arranging them in ascending order of sub1 marks. vi. Display student name whose name ends with 'h' and subject 2 marks are between 60 to 75.	Analysis / Application	6
C) a. Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R? b. Consider a B+-tree in which the maximum number of keys in a node is five. Calculate the minimum number of keys in any non-root node.	Analysis	6
Q.3 Solve Any Two of the following.		12

A) Explain following operators in Relational Algebra.	Understanding	6
	g /	
	Application	
1. Selection (σ)		
2. Projection (π)		
3. Union(U)		
4. Set Difference (-)		
5. Rename (ρ)		
6. Cartesian Product (X)		
B) Explain various Data Definition Commands in details with syntax.	Understanding	6
	g /	
	Application	
C) Explain Order by, Group by and Having Clauses with example.	Understanding	6
	g /	
	Application	
Q.4 Solve Any Two of the following.		12
A) What is RAID? Explain any five RAID levels with a neat diagram.	Synthesis	6
B) Explain remote backup systems.	Understanding	
	g /	
	Application	
C) Draw a B-tree of minimum degree as 3 by inserting the following data (10, 20, 30, 40, 50, 60, 70, 80 and 90)	Application	6
Q. 5 Solve Any Two of the following.		12
A) State and Explain 1 NF and 2 NF with example.	Understanding	6
	g /	
	Application	
B) Define transaction. List and explain ACID properties with suitable example.	Understanding	6
	g /	
	Application	
C) Explain steps in query processing with a neat diagram.	Synthesis	6
*** End ***		